2.4.4 Explicit and Implicit Methods

All of the methods mentioned so far are said to be *explicit* discretizations, since producing the next approximation point on a trajectory does not require solving any implicit equations defined in terms of the previous one. The Verlet method is not implicit even though $\boldsymbol{Q}$ appears on the right in the second equation, since it is given in an explicit way in terms of $\boldsymbol{q}$ and $\boldsymbol{p}$. Implicitness adds another layer in both analysis and numerical implementation, which, however, in certain applications, is readily justified.

Example 2.6 The *Backward Euler* method,

$$z_{n+1} = z_n + h\boldsymbol{f}(z_{n+1}),$$

is an example of an implicit method. The calculation of a timestep involves solving a system of equations of the form

$$\boldsymbol{g}(\boldsymbol{w}) \equiv \boldsymbol{w} - z_n - h\boldsymbol{f}(\boldsymbol{w}) = 0.$$

The map $\mathscr{G}_t$ is defined implicitly by the equation

$$\mathscr{G}_h(z) = z + h\boldsymbol{f}(\mathscr{G}_h(z)).$$

An implicit method will typically result in a system of nonlinear equations of the form

$$\boldsymbol{g}(z_{n+1}) = \boldsymbol{\tau}_n, \tag{2.25}$$

which will need to be solved at each timestep. The right hand side $\boldsymbol{\tau}_n$ is a vector that depends on the previous time-step z_n, perhaps in a complicated way. We may assume the number of equations represented by (2.25) is equal to the dimension of the space where z is defined, so we have a square nonlinear system. Typically $\boldsymbol{g}$ will depend on the stepsize and coefficients of the method and will have the property that for h sufficiently small, the solution is uniquely defined and is continuously defined in terms of $\boldsymbol{\tau}_n$, that is the mapping $\boldsymbol{g}$ has a bounded and smooth inverse.

Solving the system may then proceed, from an initial guess $z_{n+1}^{(0)}$ by use of Newton's method:

$$z_{n+1}^{(k+1)} = z_{n+1}^{(k)} - \left[\boldsymbol{J}^{(k)}\right]^{-1} (\boldsymbol{g}(z_n^{(k)}) - \boldsymbol{\tau}_n),$$

where $\boldsymbol{J}^{(k)}$ is the Jacobian matrix of the mapping $\boldsymbol{g}$ evaluated at $z_n^{(k)}$, or else an approximation of this Jacobian (assumed to be nonsingular due to the invertibility